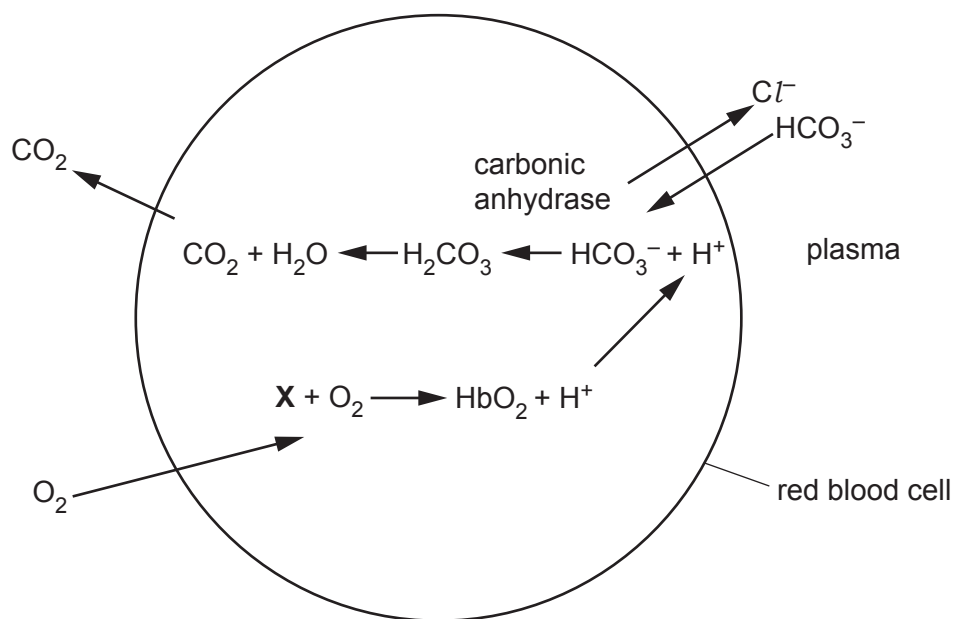


- 6 Fig. 6.1 shows some of the events that occur when a red blood cell flows through a capillary in the lungs.



**Fig. 6.1**

- (a) State why there are transport proteins in the membranes of red blood cells to allow the movement of hydrogencarbonate ions ( $\text{HCO}_3^-$ ) and chloride ions ( $\text{Cl}^-$ ).

.....  
 .....  
 ..... [1]

- (b) Fig. 6.1 shows that chloride ions move out of the red blood cells.

Explain why this movement is necessary when red blood cells flow through capillaries in the lungs.

.....  
 .....  
 ..... [1]

- (c) State why carbon dioxide molecules diffuse from the red blood cells into the plasma.

.....  
 ..... [1]

- (d) State the name of the compound indicated by **X**.

.....  
 ..... [1]

[Total: 4]